$$\hat{\rho}_{\mathcal{E}_i} = \frac{1}{Z_i} \mathrm{e}^{-\omega_i \hat{a}_i^\dagger \hat{a}_i / k_\mathrm{B} T} \equiv \frac{1}{Z_i} \mathrm{e}^{-\hat{H}_{\mathcal{E}_i} / k_\mathrm{B} T}. \tag{5.111}$$